Toward Solving the Wire-Stiffening Problem

Guidewire-vessel mechanics on SE(3), and anatomy-informed neural networks

David P. Stonko, MD, MS[1]; **Jeff K. Jopling, MD, MSHS**[2]; **Caitlin W. Hicks, MD, MS**[1]; **James H. Black III, MD**[1]

[1] Division of Vascular Surgery and Endovascular Therapy, Department of Surgery, The Johns Hopkins Hospital, Baltimore, MD
[2] Division of Acute Care Surgery, Department of Surgery, Johns Hopkins University School of Medicine, Baltimore, MD

Johns Hopkins University — Data Science and AI Institute

Johns Hopkins Medicine — The Johns Hopkins Hospital

Abstract

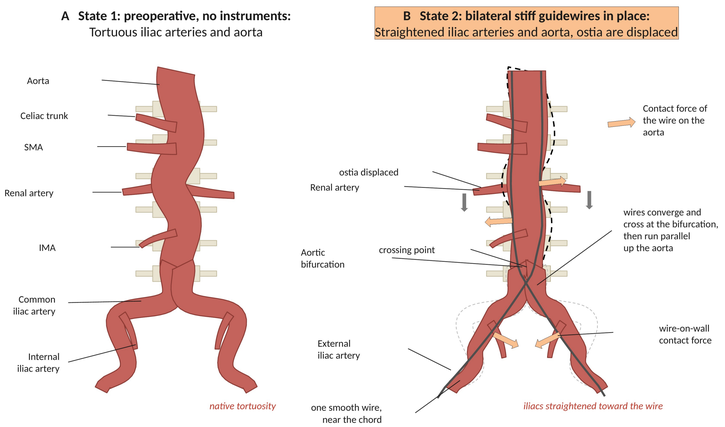
More than 130,000 aortic and aortic-valve procedures in the US each year rely on a stiff guidewire being placed into the vascular tree, over which a device travels. These procedures are planned on a CT taken before any wire is in the patient. The device is then deployed into anatomy the wire has already changed, seen through a single X-ray projection. **Can the wire-artery interaction be modeled from first principles?**

We lift the vessel centerline and the wire to curves of frames in SE(3) and rest the vessel on an elastic foundation. One inequality keeps the wire in the lumen. When the surgeon lets go, wire and vessel settle into the configuration of least energy. This model is the physics layer of **anatomy-informed neural networks**, in which anatomic knowledge enters as a hard constraint in the architecture or a soft penalty in the loss, the way physics enters a PINN.

How an aortic surgery is planned

The plan is frozen at D. By E the anatomy it was measured on no longer exists, and E is the only view we ever get of the one that replaced it.

A real, de-identified case. Every measurement (A to D) is made by hand on a CT with nothing in the patient. The operation (E) happens in anatomy the wires have already changed, seen through one projection.

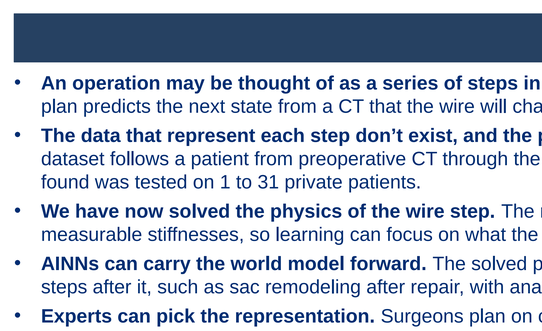


The plan is a forecast. Stiff wires straighten the iliac arteries, shift the aorta and move the branch ostia by 5 to 8 mm on average and up to 14.5 mm (Maurel 2014, Jansen 2021). Iliac tortuosity falls by 60 to 79% on the wire side (Emendi 2023). Every landmark placed in State 1 has to line up with its branch in State 2.

The wire and the vessel settle into one energy minimum

The centerline is not the wire path, and the wire must stay inside the vessel.

$$\text{gap}(s) = \left[R_{\text{lumen}}(\sigma^*(s)) - r_w\right] - d_c(s) \;\geq\; 0, \qquad d_c(s) = \min_{\sigma} \|p_w(s) - p_v(\sigma)\|$$



Definitions	
s	arc length along the wire
σ	arc length along the vessel
$p_w(s)$, $p_v(\sigma)$	points on each curve
$\sigma^*(s)$	vessel point nearest the wire
$d_c(s)$	that nearest distance
R_{lumen}, r_w	lumen and wire radii
gap(s)	clearance, must stay ≥ 0

$$\text{gap} \geq 0, \qquad \lambda \geq 0, \qquad \lambda \cdot \text{gap} = 0$$

One inequality couples wire and vessel. No springs and no penalties. The contact forces are its Lagrange multipliers.

The equilibrium minimizes the total energy of the system: the wire, vessel and tissue

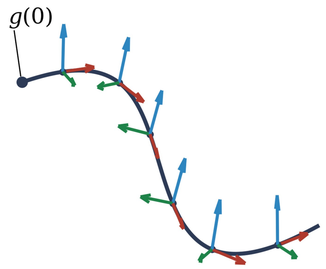
$$\mathcal{E}(w,u) = \tfrac{1}{2}\int_0^{\ell} B\,|w''|^2\,ds + V(u) + \tfrac{1}{2}\int_0^{L_v} k\,|u|^2\,d\sigma, \qquad w(s) \in \mathcal{K}(P+u) \;\text{ for all } s$$

$$H = N + \tfrac{1}{2}B\kappa^2 \;\text{ is conserved. A free tip gives } H = 0, \;\text{ so } N = -\tfrac{1}{2}B\kappa^2$$

$$\lambda = -B\left(\kappa'' + \tfrac{1}{2}\kappa^3\right) \geq 0 \;\text{ on contact, so on the inner wall } \kappa'' \leq -\tfrac{1}{2}\kappa^3 \text{: the curvature peaks}$$

w the wire, u the vessel displacement, B the wire stiffness, V the vessel bending energy, k the surrounding tissue. The minimizer is State 2, and it exists. With the tip free, the sheath holds no axial force and the wall load comes from bending alone.

The state: centerline and wire lifted into SE(3)



$$g(s) = (R(s),\, p(s)) \in SE(3)$$

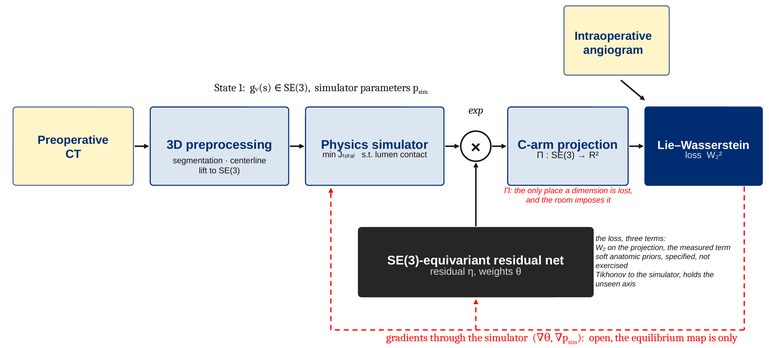
$$R(s) = [\,t \,|\, m_1 \,|\, m_2\,]$$

$$\xi_i = \tfrac{1}{\Delta s_i}\log(g_i^{-1} g_{i+1}) \in \mathfrak{se}(3)$$

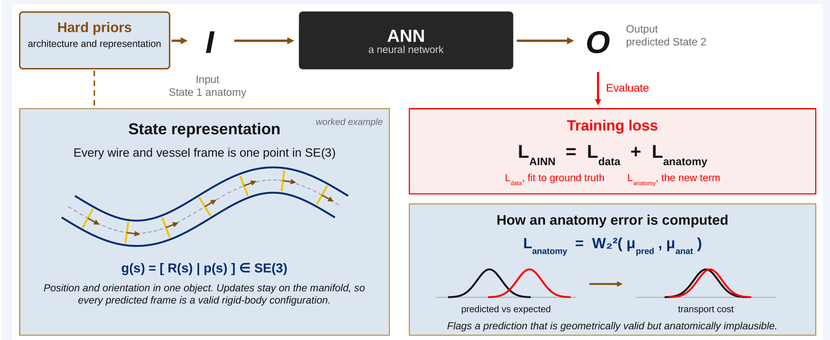
tangent t
Bishop normal m_1
Bishop normal m_2
centerline in $\mathbb{R}^3$
frame origin $p(s)$

Each point carries a position p(s) and an orientation R(s), one object **g(s) ∈ SE(3).** Position on the CT table drops out. Two vessels close together are told apart by orientation. Frames follow the curve by parallel transport.

Anatomy-informed neural networks: toward a vascular world model



One step of a world model. The physics we solved maps State 1 to State 2, and an SE(3)-equivariant residual network corrects what the physics leaves out, so every predicted frame is still a rigid-body pose. Later steps, such as remodeling after repair, chain on the same way.



Hard priors (a valid frame at every point, a continuous vessel) live in the architecture and cannot be violated. **Soft priors** are penalties in the loss: a transplant renal artery off the iliac is unusual, not impossible. Like a PINN residual, the anatomy term needs no ground truth.

Conclusions

- **An operation may be thought of as a series of steps in a world model.** The state is the anatomy, the action is the procedure, and the plan predicts the next state from a CT that the wire will change.
- **The data that represent each step don't exist, and the previously described physics that connects them is incomplete.** No public dataset follows a patient from preoperative CT through the plan, the procedure and follow-up. Every published wire-deformation model we found was tested on 1 to 31 private patients.
- **We have now solved the physics of the wire step.** The next state is the least-energy configuration. It exists and depends on a few measurable stiffnesses, so learning can focus on what the physics leaves out.
- **AINNs can carry the world model forward.** The solved physics connects State 1 to State 2. Anatomy-informed networks can learn the steps after it, such as sac remodeling after repair, with anatomy enforced at every step.
- **Experts can pick the representation.** Surgeons plan on centerlines with an orientation at every point. Curves of frames in SE(3) make this exact and extend to the whole vascular tree.
- **A clinical model should know when to abstain.** Our planner declines a poor scan rather than return a bad plan.

Next: EVAR-Pair, a planned open library of complete EVAR episodes with benchmarks, then an anatomy-informed world model and an endovascular copilot.

Collaborate
Looking for an applied mathematician and ML engineers
localminimum.us/collaborate

Website
localminimum.us

Preprint
arXiv:2608.21332

Code
github.com/davidstonko
Poster made with Claude Fable 5.1 using the /jhsurgeryformat skill.